

Solenoidal Nature of $\mathbf{H}$

In an isotropic material $\mathbf{B} = \mu\mathbf{H}$. This means that $\mathbf{H}$ is always solenoidal.

- ☒ a. No
- ☐ b. Yes
- ☐ c. Yes but just in the paramagnetic and ferromagnetic materials.
- ☐ d. Yes but just in a diamagnetic material.

Definition of bulk amperian current

The bulk density of amperian currents $\mathbf{j}_m$ in a material with magnetization vector $\mathbf{M}$, is equal to

- a. $\mathbf{j}_m = \text{rot } \mathbf{M}$
- b. $\mathbf{j}_m = \mathbf{M} \times \mathbf{n}$
- c. $\mathbf{j}_m = \text{div } \mathbf{M}$
- d. $\mathbf{j}_m = \text{grad } \mathbf{M}$

Force per unit length between parallel wires

Two straight wires of indefinite length, placed in vacuum parallel to each other, are crossed by currents having the same intensity $I = 5 \text{ A}$ and towards concordance. If the two wires are $d = 10 \text{ cm}$ from each other, how much is the force per unit of length that they exert on each other in module? Is it a repulsive or attractive force?

a. $F = 5 \cdot 10^{-5} \text{ N / m}$, attractiveness

b. $F = 5 \cdot 10^{-5} \text{ N / m}$, repulsive

c. $F = 5 \cdot 10^{-4} \text{ N / m}$, attractiveness

d. $F = 5 \cdot 10^{-4} \text{ N / m}$, repulsive

Validity of Biot-Savart law approximation

When is the Biot-Savart law applicable (in the context of the standard infinite wire approximation)?

- a. Never
- ☒ b. When the distance from the wire is much less than the length of the wire
- c. Always
- d. When the distance from the wire is equal to the length of the wire
- e. When the distance from the wire is much greater than the length of the wire

General formula for Biot-Savart law

The magnetic field $\mathbf{B}(\mathbf{r})$ generated by a threadlike circuit s' , travelled by a current I , in stationary conditions, in a generic point of space identified by the vector $\mathbf{r}$ is ($\mathbf{r}'$ identifies a point of the circuit s'):

a. $\mathbf{B}(\mathbf{r}) = (I/4\pi) \oint d\mathbf{s}' \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3$

b. $\mathbf{B}(\mathbf{r}) = (\mu_0 I/4\pi) \oint d\mathbf{s}' \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3$

c. $\mathbf{B}(\mathbf{r}) = (I/4\pi\mu_0) \oint d\mathbf{s}' \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3$

d. $\mathbf{B}(\mathbf{r}) = (\mu_0 I/4\pi) \oint d\mathbf{s}'/|\mathbf{r} - \mathbf{r}'|^3$

e. $\mathbf{B}(\mathbf{r}) = (\mu_0 I/4\pi) \oint d\mathbf{s}' \times (\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^2$

Particle motion parallel to magnetic field

A particle of charge q and mass m moves in a uniform rectilinear motion with speed $\vec{v} = v_0 \hat{\mathbf{i}}$ until it enters a region where a uniform magnetic field $\vec{B} = B_0 \hat{\mathbf{i}}$ is present. What happen?

- a. The particle immediately stops
- ☒ b. Nothing
- c. The particle moves along a helix
- d. The particle moves along a circle
- e. The particle moves along a circumference with uniform circular motion

Applications of the Hall effect

The Hall effect can be used:

- a. to determine the dielectric strength of an insulator:
- b. to determine the mobility of the ions in a generic metal
- ☒ c. to measure a magnetic field
- d. to determine the intensity of an electromagnetic wave

Force on a closed circuit in uniform $\mathbf{B}$

The second law of Laplace provides that the force applied on a closed circuit crossed by a stationary current I on which a uniform magnetic field $\mathbf{B}$ acts:

- a. dependent only on the area enclosed by the circuit
- b. non-zero if field $\mathbf{B}$ is parallel to the circuit
- c. different from zero if field $\mathbf{B}$ is orthogonal to the circuit
- ☒ d. always zero

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Magnetic field from vector potential

In a region of empty space there exists a vector potential $\mathbf{A}(x, y, z, t) = ax\mathbf{i} + by\mathbf{j} + ctz\mathbf{k}$ (where a, b , and c are constants of suitable size). How much is the associated magnetic field $\mathbf{B}$?

☒ a. $\mathbf{B} = -ct\mathbf{j}$

b. $\mathbf{B} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$

c. $\mathbf{B} = -ct\mathbf{j}$

d. $\mathbf{B} = at\mathbf{i} + b\mathbf{j} + c\mathbf{k}$

Work done by Lorentz force

An electron of charge e and velocity of modulo v completes a semicircle of radius R due to a magnetic field of modulo B , placed perpendicular. How much is the work done by the Lorentz force worth?

- a. $W = 0$ J
- b. $W = \pi evBR/2$ J
- c. $W = \pi evBR$ J
- d. $W = 2\pi evBR$ J

Conservative condition for magnetic field

The magnetic field $\mathbf{B}(x, y, z) = ax\mathbf{i} - by\mathbf{j} + 5cx\mathbf{k}$ (where a , b , and c are constants of suitable size) is conservative?

- a. Just when $b = 0$
- ☒ b. Just when $c = 0$
- c. Always
- d. Just when $a = 0$

Gauss's law for magnetic field

Gauss's law for the magnetic field $\mathbf{B}$ necessarily provides that:

- a. The flux of $\mathbf{B}$ is always zero if the magnetization of the material is negligible
- b. The flow of $\mathbf{B}$ is always zero if there are no amperian currents
- c. The flow of $\mathbf{B}$ across a surface is always zero
- d. There are no magnetic monopoles

Torque on a circular winding

A circular winding with radius $R = 7$ cm consists of $N = 300$ turns crossed by a current $i = 15$ A. The axis of the winding, oriented in the direction that sees the circular current in an anticlockwise direction, is parallel to the unit vector $\mathbf{i}$. The winding is in a magnetic field $\mathbf{B} = 0.3\mathbf{i} + 0.4\mathbf{j} + 0.5\mathbf{k}$ (Tesla). Calculate the value of the modulus of the moment $\mathbf{M}$ (torque) to which the winding is subjected.

a. $M = 11.1$ Nm

b. $M = 22.2$ Nm

c. $M = 55.5$ Nm

d. $M = 44.4$ Nm

Ampère's Law and Concatenated Current

Can the integral of the magnetic field $\mathbf{B}$ along a closed line L be zero if there exists a current I concatenated with L ?

- ☒ a. Yes if the current I is connected to L twice in the opposite direction
- b. Yes if the current I is connected to L twice in the same direction
- c. Yes if the field $\mathbf{B}$ is static
- d. It's impossible
- e. Yes if the space around L is filled with a diamagnetic material

Motion of a particle in electromagnetic fields

In a region of space there is an electric field $\mathbf{E} = 4abz\mathbf{i}$ and a magnetic field $\mathbf{B} = 2b\mathbf{j}$ where a and b are constants of suitable size. What is the system of differential equations which, with suitable boundary conditions, describes the trajectory of a point particle of charge q and mass m ? (Neglecting gravity)

a. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = 4a(q/m)z(t); \quad \frac{d^2z(t)}{dt^2} = 0$

b. $\frac{d^2x(t)}{dt^2} = 2b(q/m) \left[2az(t) - \frac{dz(t)}{dt} \right]; \quad \frac{d^2y(t)}{dt^2} = 0; \quad \frac{d^2z(t)}{dt^2} = 2b(q/m) \frac{dx(t)}{dt}$

c. $\frac{d^2x(t)}{dt^2} = -2b(q/m)z(t) \frac{dx(t)}{dt}; \quad \frac{d^2y(t)}{dt^2} = 4ab(q/m)z(t); \quad \frac{d^2z(t)}{dt^2} = 2b(q/m)x(t) \frac{dx(t)}{dt}$

d. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = 2(q/m)(2az(t) + b\dot{x}(t)); \quad \frac{d^2z(t)}{dt^2} = 0$

Torque on a square coil

A rigid square coil of side $a = 20$ cm is crossed by a current $I = 2$ A and is placed in a magnetic field of modulus $B = 0.03$ T whose direction forms an angle $\vartheta = 30^\circ$ with the normal on the plain on which the coil lies. Determine the modulus of torque M acting on the loop.

- a. $M = 0$ Nm
- b. $M = 2.4 \cdot 10^{-4}$ Nm
- c. $M = 2.4 \cdot 10^{-2}$ Nm
- d. $M = 1.2 \cdot 10^{-2}$ Nm
- ☒ e. $M = 1.2 \cdot 10^{-3}$ Nm

Magnetic field between parallel wires

Two straight parallel wires of indefinite length, placed in the empty space, are crossed by two concordant currents I_1 and I_2 . Determine the strength B of the field placed in the middle of the straight line joining the two wires, separated by a distance d .

a. $B = \mu_0 |I_1 - I_2| / \pi d$

b. $B = \mu_0 |I_1 - I_2| / 2\pi d$

c. $B = \mu_0 |I_1 + I_2| / \pi d$

d. $B = \mu_0 |I_1 + I_2| / 2\pi d$

e. $B = \mu_0 |I_1 - I_2| / \pi d^2$

Magnetic field inside a current-carrying cylinder

An indefinite cylinder is traversed by a current parallel to the axis of the cylinder itself with uniform and stationary current density j . The field $\mathbf{B}$ inside the cylinder:

- a. is always null
- ☒ b. grows linearly as the distance r from the center increases
- c. is always constant
- d. is not known. The data of problem are not sufficient to answer.

Angular velocity of a proton in a magnetic field

A proton moves in a region of space in which there is a uniform magnetic field having a modulus of $B = 0.05 \text{ T}$ describing circular orbits of radius $r = 2 \text{ m}$. Find the magnitude of the angular velocity ω of the proton.

- a. The angular velocity ω cannot be determined.
- b. $\omega = 2.39 \cdot 10^6 \text{ rad/s}$
- ☒ c. $\omega = 4.79 \cdot 10^6 \text{ rad/s}$
- d. $\omega = 9.58 \cdot 10^6 \text{ rad/s}$

Magnetic field of a moving charge

The magnetic field produced by a charge in motion is:

- a. directed parallel to the electric field generated by charge
- b. direct parallel to the direction of the line joining the charge with the point at which you want to measure the magnetic field
- ☒ c. directed perpendicular to the speed of the charge
- d. inversely proportional to the distance between the charge and the point at which you want to measure the magnetic field

Magnetic field inside a long solenoid

If the current flowing in a long solenoid increases of four times what happen to magnetic field inside the solenoid:

- a. it remains constant
- ☒ b. it quadruples
- c. it doubles
- d. it is halved

Magnetic field from vector potential

In a region of empty space there exists a potential vector $\mathbf{A}(x, y, z, t) = ax t \mathbf{u}_x + by \mathbf{u}_y + cx \mathbf{u}_z$ (where a, b , and c are constants of suitable size). What is the associated magnetic field $\mathbf{B}$?

Solenoidal nature of the magnetic field

Under what conditions the magnetic field vector $\mathbf{B}$ is solenoidal ($\text{div } \mathbf{B} = 0$)?

- a. Under steady current conditions
- b. When there are no charges in motion
- c. In the empty space
- ☒ d. Always

Trajectory of a charged particle in a magnetic field

A charged particle q and mass m enters a zone in which there is a magnetic field $\mathbf{B}$ having components $(0, B_0, 0)$ (with respect to a Cartesian orthogonal reference system) with a velocity $\mathbf{v}$ having components $(0, v_1, v_2)$. What trajectory does the particle describe and what is its radius?

- a. helix, $r = \frac{mv_2}{qB_0}$
- b. circumference, $r = \frac{mv_2}{qB_0}$
- c. helix, $r = \frac{mv_1}{qB_0}$
- d. circumference, $r = \frac{mv_1}{qB_0}$

Magnetic field of a moving point charge

The magnetic field generated at a point P by a moving (non-relativistic) charge

- a. does not depend on the direction of motion of the charge
- b. is inversely proportional to the distance between the charge and the point P
- c. does not depend on the magnitude of the charge rate
- ☒ d. is inversely proportional to the square of the distance between the charge and the point P

Magnetic energy of a current loop in an external field

How much is the magnetic energy possessed by a coil square of side $l = 20$ cm, traversed by a current $i = 2$ A and immersed in a magnetic field $\mathbf{B} = 0.1$ T directed antiparallel (the same direction but opposite way journey) to the normal to the plane of the spire.

a. $U_m = 0$ J

☒ b. $U_m = 0.008$ J

c. $U_m = 0.016$ J

d. $U_m = -0.016$ J

Motion of a charged particle in a magnetic field

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Gauss's law for magnetism

The equation $\nabla \cdot \mathbf{B} = 0$ is:

- ☒ a. always verified
- ☐ b. verified only in the areas of space where there are electric charges in motion
- ☐ c. verified only in the areas of the space where the magnetic field is null
- ☐ d. occurred only in the areas of space where there are no electric currents

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Mutual Inductance of a Wire-Solenoid System

A very long rectilinear wire is traversed by a electric current of variable intensity $I(t) = I_0 \sin(\omega t)$. A solenoid consisting of N square coils of side a is arranged as in figure at distance b from the wire. What is the coefficient of mutual inductance M of this system?

a. $M = \frac{\mu_0 N a \omega}{2\pi} \ln \left[\frac{a+b}{b} \right]$

☒ b. $M = \frac{\mu_0 N a}{2\pi} \ln \left[\frac{a+b}{b} \right]$

c. $M = \mu_0 N a / 2\pi$

d. $M = \frac{\mu_0 N a \omega I_0}{2\pi} \ln \left[\frac{a+b}{b} \right]$

e. $M = \frac{\mu_0 N a}{2\pi} \ln \left[\frac{a+b}{a} \right]$

Condition for Magnetic Field to be Solenoidal and Conservative

Under what conditions is the vector magnetic field $\mathbf{B}$ solenoidal and conservative?

- a. Only in steady state.
- ☒ b. When $\text{rot } \mathbf{B} = 0$.
- c. When there are no charges.
- d. Always.